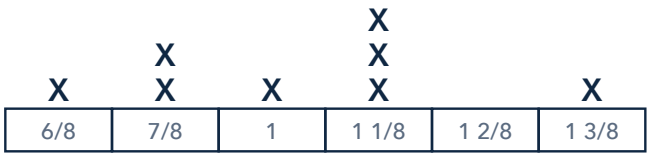
Line Plots with Fractions

Put an X for each one. Then add and take away fractions.

8 bugs, measured to the eighth of an inch

6/8 in	7/8 in	7/8 in	1 in
1 1/8 in	1 1/8 in	1 1/8 in	1 3/8 in

Bug lengths (inches)



Questions you can answer

Longest minus shortest: $1\frac{3}{8} - \frac{6}{8} = \frac{5}{8}$ in

All the 1 1/8 in bugs end to end: $3 \times 1\frac{1}{8} = 3\frac{3}{8}$ in

How many are shorter than 1 in? 3 bugs

Your turn!

Longest 1 2/8 in, shortest 5/8 in.

$$1\frac{2}{8} - \frac{5}{8} = \frac{\boxed{5}}{8}$$

For teachers

Goal: X charts (line plots) of measurements in halves, quarters and eighths, then add and subtract the fractions to answer questions (4.MD.4). Say "Find the biggest and smallest X first."

Line Plots with Fractions: Practice

One X for each measurement.

Part 1 Make the X chart Put an X above the length for each one.

1)

Ribbon pieces

5/8 in 7/8 in

6/8 in 7/8 in

6/8 in 1 1/8 in

7/8 in 1 1/8 in

Make the X chart (inches)



2)

Leaves

1 1/8 in 1 4/8 in

1 2/8 in 1 4/8 in

1 2/8 in 1 5/8 in

1 4/8 in 1 6/8 in

Make the X chart (inches)



3)

Snails

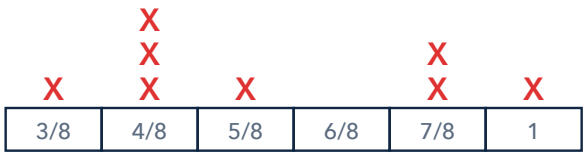
3/8 in 5/8 in

4/8 in 7/8 in

4/8 in 7/8 in

4/8 in 1 in

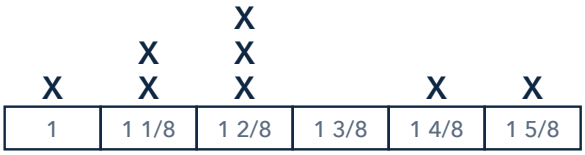
Make the X chart (inches)



Line Plots with Fractions: Practice (continued)

Part 2 Read the X chart Answer the questions.

4) **Feathers (inches)**



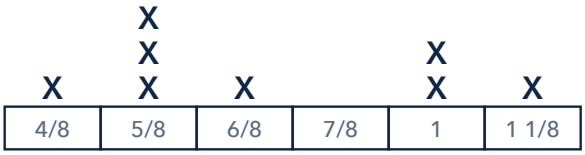
How many are 1 1/8 in? **2**

Longest minus shortest? **5/8**

All the 1 2/8 in ones together? **3 6/8**

How many in all? **8**

5) **Shells (inches)**



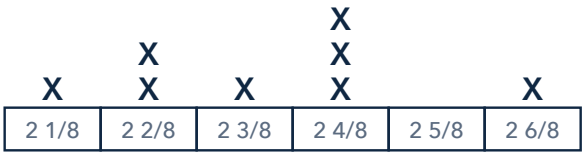
How many are 4/8 in? **1**

Longest minus shortest? **5/8**

All the 5/8 in ones together? **1 7/8**

How many in all? **8**

6) **Pencils (inches)**



How many are 2 2/8 in? **2**

Longest minus shortest? **5/8**

All the 2 4/8 in ones together? **7 4/8**

How many in all? **8**

Line Plots with Fractions: Practice (continued)

Part 3 Add the X's No help this time. Same length, many times.

7) $3 \times \frac{3}{8} = \frac{9}{8}$

8) $4 \times \frac{5}{8} = \frac{20}{8}$

9) $5 \times \frac{3}{8} = \frac{15}{8}$

10) $5 \times \frac{2}{8} = \frac{10}{8}$

11) $3 \times \frac{6}{8} = \frac{18}{8}$

12) $6 \times \frac{1}{8} = \frac{6}{8}$

13) $2 \times \frac{5}{8} = \frac{10}{8}$

14) $3 \times \frac{7}{8} = \frac{21}{8}$

Part 4 How far apart? No help this time. Longest minus shortest.

15) $1\frac{5}{8} - \frac{6}{8} = \frac{7}{8}$

16) $1\frac{7}{8} - \frac{4}{8} = 1\frac{3}{8}$

17) $1\frac{3}{8} - \frac{4}{8} = \frac{7}{8}$

18) $1\frac{6}{8} - 1\frac{1}{8} = \frac{5}{8}$

Line Plots with Fractions: Practice (continued)

Part 4 How far apart? (continued)

19) $2\frac{4}{8} - \frac{7}{8} = \boxed{1}\frac{\boxed{5}}{8}$

20) $1\frac{2}{8} - \frac{5}{8} = \frac{\boxed{5}}{8}$

21) $2\frac{1}{8} - \frac{2}{8} = \boxed{1}\frac{\boxed{7}}{8}$

22) $1\frac{4}{8} - 1\frac{2}{8} = \frac{\boxed{2}}{8}$

Part 5 Put them together No help this time. Two pieces end to end.

23) $1\frac{1}{8} + \frac{4}{8} = \boxed{1}\frac{\boxed{5}}{8}$

24) $\frac{5}{8} + \frac{6}{8} = \boxed{1}\frac{\boxed{3}}{8}$

25) $1\frac{2}{8} + 1\frac{3}{8} = \boxed{2}\frac{\boxed{5}}{8}$

26) $\frac{3}{8} + \frac{2}{8} = \frac{\boxed{5}}{8}$

27) $1\frac{4}{8} + \frac{7}{8} = \boxed{2}\frac{\boxed{3}}{8}$

28) $\frac{6}{8} + 1\frac{1}{8} = \boxed{1}\frac{\boxed{7}}{8}$

29) $2\frac{1}{8} + \frac{3}{8} = \boxed{2}\frac{\boxed{4}}{8}$

30) $\frac{1}{8} + \frac{5}{8} = \frac{\boxed{6}}{8}$

Line Plots with Fractions: Practice (continued)

Part 6 Add the eighths

31) $\frac{4}{8} + \frac{4}{8} = \frac{8}{8}$

32) $\frac{7}{8} + \frac{7}{8} = \frac{14}{8}$

33) $\frac{5}{8} + \frac{1}{8} = \frac{6}{8}$

34) $\frac{6}{8} + \frac{4}{8} = \frac{10}{8}$

35) $\frac{2}{8} + \frac{5}{8} = \frac{7}{8}$

36) $\frac{5}{8} + \frac{4}{8} = \frac{9}{8}$

37) $\frac{2}{8} + \frac{2}{8} = \frac{4}{8}$

38) $\frac{6}{8} + \frac{3}{8} = \frac{9}{8}$

39) $\frac{2}{8} + \frac{1}{8} = \frac{3}{8}$

40) $\frac{6}{8} + \frac{2}{8} = \frac{8}{8}$

41) $\frac{1}{8} + \frac{2}{8} = \frac{3}{8}$

42) $\frac{7}{8} + \frac{2}{8} = \frac{9}{8}$

Line Plots with Fractions: Practice (continued)

Part 6 Add the eighths (continued)

43) $\frac{6}{8} + \frac{5}{8} = \frac{11}{8}$

44) $\frac{7}{8} + \frac{6}{8} = \frac{13}{8}$

45) $\frac{3}{8} + \frac{5}{8} = \frac{8}{8}$

46) $\frac{1}{8} + \frac{1}{8} = \frac{2}{8}$

47) $\frac{7}{8} + \frac{5}{8} = \frac{12}{8}$

48) $\frac{5}{8} + \frac{7}{8} = \frac{12}{8}$

Part 7 Times eighths

49) $5 \times \frac{5}{8} = \frac{25}{8}$

50) $2 \times \frac{6}{8} = \frac{12}{8}$

51) $6 \times \frac{7}{8} = \frac{42}{8}$

52) $2 \times \frac{3}{8} = \frac{6}{8}$

53) $4 \times \frac{6}{8} = \frac{24}{8}$

54) $3 \times \frac{4}{8} = \frac{12}{8}$

Line Plots with Fractions: Practice (continued)

Part 7 Times eighths (continued)

55) $6 \times \frac{5}{8} = \frac{30}{8}$

56) $5 \times \frac{4}{8} = \frac{20}{8}$

57) $6 \times \frac{4}{8} = \frac{24}{8}$

58) $4 \times \frac{1}{8} = \frac{4}{8}$

59) $6 \times \frac{6}{8} = \frac{36}{8}$

60) $4 \times \frac{2}{8} = \frac{8}{8}$

61) $5 \times \frac{1}{8} = \frac{5}{8}$

62) $4 \times \frac{4}{8} = \frac{16}{8}$

63) $5 \times \frac{7}{8} = \frac{35}{8}$

64) $4 \times \frac{7}{8} = \frac{28}{8}$

Line Plots with Fractions: Practice (continued)

Part 8 Remember? Problems from earlier packets.

65) area: 13 × 6 = 78 sq m

66) 5 ft = 60 in

67) 0.8 > 0.75

68) $5 \times \frac{3}{10} = \frac{15}{10}$

69) $3\frac{1}{8} - 1\frac{2}{8} = 1\frac{7}{8}$

70) 75 × 12 = 900

Part 9 Word problems Use eighths.

71) Two worms are 1 4/8 in and 7/8 in long. How much longer is the first one?

Number sentence 1 4/8 − 7/8 = 5/8

Answer 5/8 in

Draw or show your work

72) Four beads are each 3/8 in long. How long are they in a row?

Number sentence 4 × 3/8 = 12/8 = 1 4/8

Answer 1 4/8 in

Draw or show your work

73) Ana's X chart shows 3 X's at 5/8 in and 2 X's at 7/8 in. How many things did she measure?

Number sentence 3 + 2 = 5

Answer 5 things

Draw or show your work

Line Plots with Fractions: Practice (continued)

Part 9 Word problems (continued)

74) A ribbon is 2 1/8 in. Another is 1 4/8 in. How long are they together?

Number sentence 2 1/8 + 1 4/8 = 3 5/8

Answer 3 5/8 in

Draw or show your work

75) Challenge: The longest leaf is 1 7/8 in and the shortest is 6/8 in. How far apart are they?

Number sentence 1 7/8 − 6/8 = 1 1/8

Answer 1 1/8 in

Draw or show your work

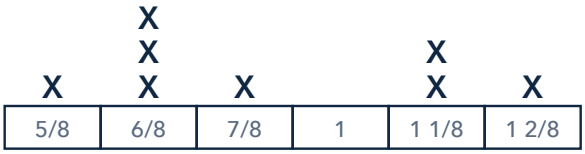
Check-In: Line Plots with Fractions

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Read the X chart

1) **Sticks (inches)**



How many are 6/8 in? 3

Longest minus shortest? 5/8

All the 6/8 in ones together? 2 2/8

How many in all? 8

Part 2 Add the X's

2) $6 \times \frac{2}{8} = \frac{12}{8}$

3) $2 \times \frac{1}{8} = \frac{2}{8}$

4) $3 \times \frac{5}{8} = \frac{15}{8}$

Part 3 How far apart?

5) $1\frac{1}{8} - \frac{4}{8} = \frac{5}{8}$

6) $1\frac{7}{8} - \frac{3}{8} = 1\frac{4}{8}$

7) $1\frac{1}{8} - \frac{2}{8} = \frac{7}{8}$